

Autodesk® Scaleform®

Scaleform 4.3

/

(Scaleform 4.3 Mobile Game Kit)

,

Scaleform

Nate Mitchell

1.01

2012 6 5

Autodesk® Scaleform® 4.3

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1		Error! Bookmark not defined.
1.1	Starforce Battlement	2
1.2		4
2		5
2.1		5
2.2		6
2.2.1		6
2.2.2	HUD	8
3		10
3.1	Flash	10
3.2	ActionScript	10
3.2.1		10
3.2.2	ActionScript	13
3.2.3		14

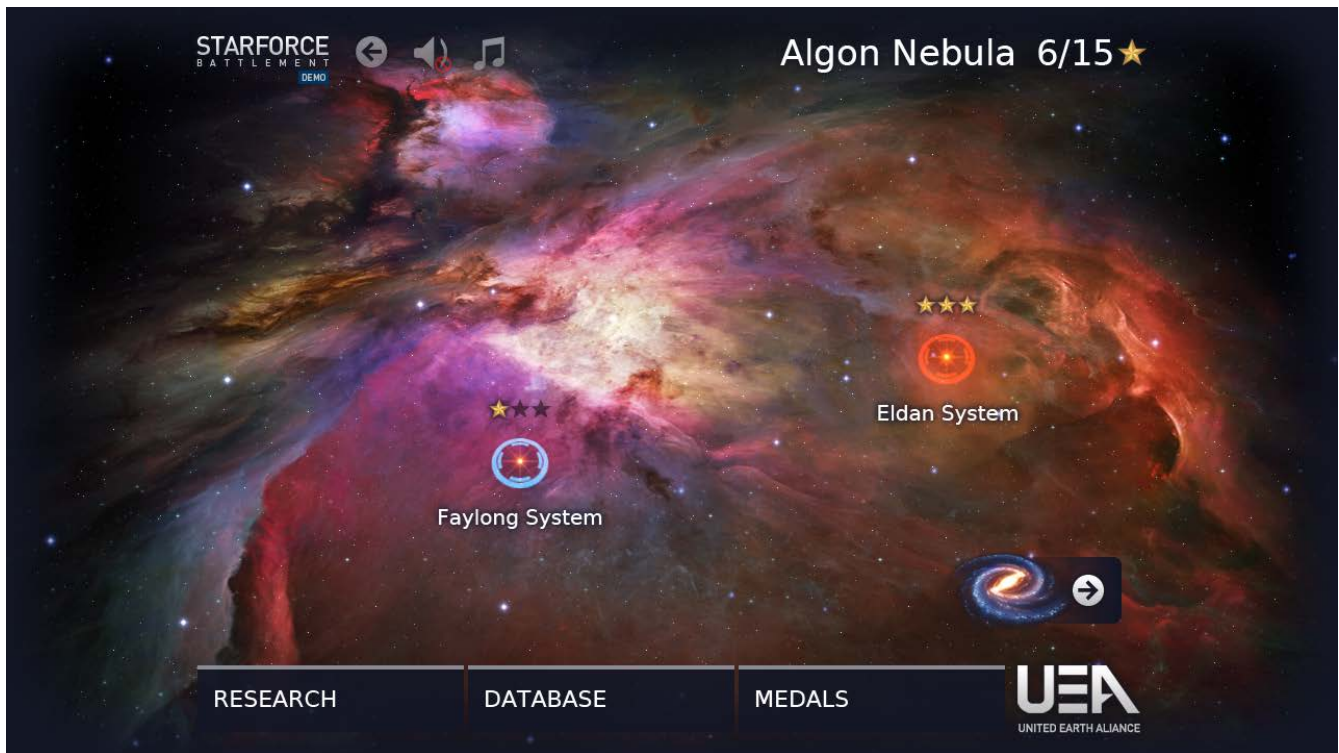
1

Scaleform
Autodesk® Scaleform®
3 , Scaleform
(PC Mac Linux iOS Android)
Starforce Battlement
(Shippable Mobile Player)
ActionScript
ActionScript 3
iOS (iOS Game Center)
Adobe® Flash® (Asset)
Scaleform
Scaleform
Web , Scaleform
Adobe Flash Player
/ Starforce Battlement
1. Flash Professional UI
2. ActionScript 3 (FlashDevelop 4
3. Scaleform Exported .SWF
4. Scaleform GPU ,

1.1 Starforce Battlement



/ (/)



2 Starforce –

,Starforce Battlement

/ ,

2012 5 , iOS App Store
 , Scaleform iOS

Starforce Battlement

iOS

1.2

Starforce Battlement

Scaleform

- 1.
2. GPU
3. ActionScript 3
- 4.
- 5.
- 6.
7. iOS
8. (Orientation Lock) / (Orientation Handling)

 ActionScript 3

- 1.
- 2.
3. iOS
- 4.

2

2.1

- *Bin/Data/AS3/Kits/StarforceTD/* - ActionScript 3 Flash
()
- *Projects/Win32 /{Msvc80, Msvc90, or Msvc10}/Gfx 4.3 SDK /* – Visual Studio ,
Windows FxPlayer Visual Studio 2005/2008/2010 ,
Starforce.
- *Projects/iPhone/Xcode4/Gfx 4.3 iPhone SDK* – Xcode 4 , iOS
(Shippable Mobile Player) , Starforce.
- *LocalApps\StarforceTD\Android* – make Scaleform,
Eclipse , Shippable Mobile Player (Android
StarforceTD

Bin/Data/AS3/Kits/ StarforceTD Windows StarforceTD.exe
Windows Scaleform SDK

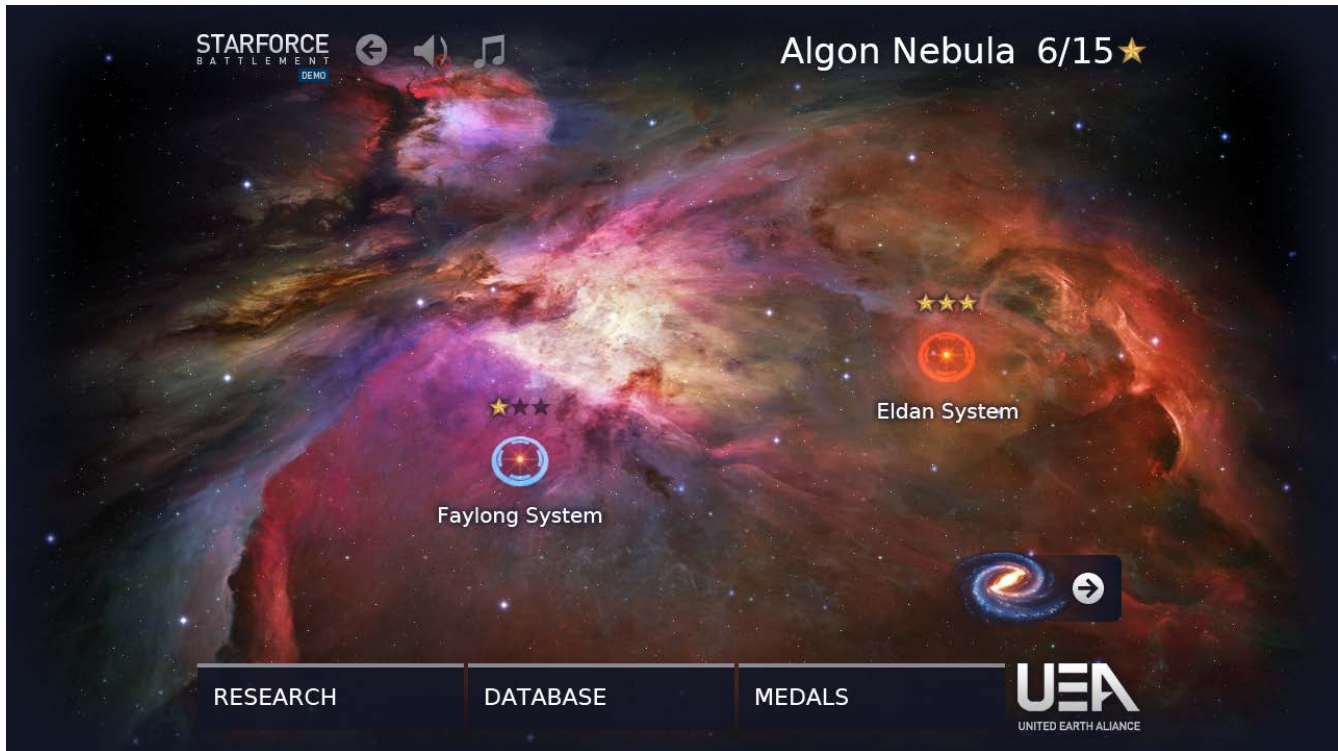
FxPlayer / FxMobilePlayer

Windows , Debugging() Working directory ()
Bin/Data/AS3/Kits/StarforceTD , Command line arguments ()
StarforceTD.swf

iOS , Xcode , iTunes SWF ()

2.2

2.2.1



3 Starforce –

StarforceTD.as StarforceTD.fla Document Class(),
 MainMainState MainMainState StarforceMenu.swf StarforceMenu.swf
, ActionScript 3

, / (galaxy/nebula)
2D ,



4 Starforce –

, (MissionDataView.as) (Mission
Data View) (Enter Mission)
X (Mission Select
View)

2.2.2 HUD



5 Starforce -

Starforce Battlement

StarforceTD.swf

StarforceTD.swf

LevelGameState

-- Rush Wave ()--

HUD

1 2 3

()

HUD

(Health)

HUD

	:	(Orbital Strike)	(Deploy Mercenaries),
HUD		,	,
	,	(	),
	,	,	
		,	
HUD		/	/

3

3.1 Flash

Battlement Flash .FLA (Bitmap) , UI Starforce 3D (UI) FLA ActionScript

Starforce Battlement FLA

- 1. **StarforceTD fla** – .FLA Starforce Battlement UI (Main Menu) / ActionScript (Loading Screen),
- 2. **StarforceMenu fla** – (Mission Select) ActionScript StarforceTD.swf , ,
- 3. **LoadingView.swf** – () ActionScript

3.2 ActionScript

3.2.1

Starforce Battlement UI ActionScript 3 , iOS Game Center , Starforce

3.2.1.1

StarforceTD.as , Flash Document Class , StarforceTD.swf (Document Classes) , Adobe Flash

3.2.1.2

```
        'GameState'(
GameState, StarforceTD
1. MenuGameState –
2. LoadingGameState –
3. LevelGameState –

StarforceTD (
MainMenuView.as StarforceMenu.swf
```

3.2.1.3

```
com.scaleform.std.menu (Package) (
) , , ActionScript 3
/ ( , )
,
LoadingGameState MenuGameState LevelGameState, LevelGameState
,
, LoadingGameState,
StarforceTD.swf , LoadingGameState
HUD,
```

3.2.1.4

```
Level.as Level.as
, HUD Level
, ( ,
, )
Level
( , )
```

Flash Professional ActionScript (Symbol),
 ActionScript , , ActionScript Starforce
 Battlement
 ActionScript , Adobe Flash

 ,TowerMech StarforceTD fla Flash Library TowerMech , ,
 com.scaleform.std.entities.TowerMech ActionScript ActionScript
 , , ,
 Level.as (, HUD , (Tower) (Unit) ,
) , , AttackEvent AttackEvents
 (Level) AttackEvent, , ,

 (, , , ,
) , , ,

3.2.1.5

com.scaleform.std.data ,
 , GameData , , (GameData (
 ,), ,GameData

 , ActionScript , XML JSON
 ,Scaleform 4.0 XML, ,
 ,

3.2.1.6

 ActionScript
 Scaleform /
 ActionScript


```

        Scaleform
        MovieView
        MultitouchInputState,
        ,
        , Starforce
        TouchEvent
MouseEvent TouchEvent MouseEvent , TouchEvent
    MouseEvent , TouchEvent.TOUCH_TAP MouseEvent.MOUSE_CLICK
    , TouchEvent.PRESS MouseEvent.MOUSE_DOWN
    TouchEvent MouseEvent , Event(
)
    , target( )
    is ,
protected function onSetRallyClick( e:Event ):void {
    var point:Point;
    if (e is MouseEvent) {
        var me:MouseEvent = e as MouseEvent;
        point = new Point( me.stageX, me.stageY );
    }
    else {
        var te:TouchEvent = e as TouchEvent;
        point = new Point( te.stageX, te.stageY );
    }
}

```

3.2.2 ActionScript

```

/ , ActionScript ActionScript' '(Package)
com.scaleform.std
1. abilities - ( )
2. controls - UI
3. core - (State Flow) /
4. data - ,
5. entities - ,
6. events - UI
7. fx - ( )
8. hud- HUD
9. levels - ,Level( )
10. loading - (Loading Screen)

```

11. menu	–	,	UI	
12. system	–			ActionScript
13. utils	–	(	)	

3.2.3

3.2.3.1 /

```
( com.scaleform.std.system.SaveSystem.as )
```

```

ActionScript      SharedObject      SharedObject      ActionScript 3
,                  ActionScript      Scaleform      SharedObject
(Primitive Type)  (Array)

```

```

SharedObject
,SharedObject      sb_player_data

```

```

// Local reference to the galaxy progress. Populated from the PlayerData.
public static var galaxyProgress:Array = null;
// Constant String for the property of the SharedObject we'll modify.
protected static const GALAXY_PROGRESS_SO_PROP:String = "galaxyProgress";
// Reference to the SharedObject once we retrieve it.
protected static var _playerDataSO:SharedObject = null;

public static function load( pd:PlayerData ):void {
    _playerDataSO = SharedObject.getLocal("sb_player_data");
    galaxyProgress = _playerDataSO.data[ GALAXY_PROGRESS_SO_PROP ];
    pd.setGalaxyProgress( galaxyProgress );
}

public static function save( pd:PlayerData ):void {
    // Write PlayerData into SharedObject.
    _playerDataSO = SharedObject.getLocal("sb_player_data");
    galaxyProgress = pd.getGalaxyProgress();
    _playerDataSO.data[ GALAXY_PROGRESS_SO_PROP ] = galaxyProgress;
    _playerDataSO.flush();
}

```

```

, SharedObject SharedObject
SharedObject

```

```

SharedObject

```

http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/net/SharedObject.html

3.2.3.2

```

(com.scaleform.std.system.SoundSystem )
SoundEvent ( .mp3) SoundEvent
stage

```

```

dispatchEvent( new SoundEvent( SoundEvent.PLAY_SOUND, true, true,
"infantry_attack.mp3", 1, false ) );

```

```

, , SoundEvent
, SoundEvent

```

```

dispatchEvent( new SoundEvent( SoundEvent.MUTE_SOUNDFX, true, true );

```